

SECTION B

6 (**short**) Consider the differential equation

$$\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 5x = 1$$

with initial conditions $x(0) = \dot{x}(0) = 0$. Show that the Laplace transform of $x(t)$ can be written in the form

$$X(s) = \frac{A}{s} + \frac{B}{[(s+1)^2 + 4]} + \frac{C(s+1)}{[(s+1)^2 + 4]}$$

where A , B and C are constants. Hence find $x(t)$. [10]

7 (**short**) A linear system has impulse response $e^{-t} - e^{-2t}$ for $t \geq 0$. Find its response to the input

$$x(t) = \begin{cases} 0 & t < 0 \\ e^{-t} & t \geq 0 \end{cases}$$

Is this a first or second order system? Justify your answer. [10]

8 (**short**) A tennis match is contested between two players, A and B. The match comprises a number of *sets*. The winner of the match is the first player to win three sets. The probability that player A wins any one set is a .

Find an expression for the probability P that player A wins the match. Verify that the value of P is as you would expect for three distinct values of a . [10]